Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

BCA Degree Examination – APRIL 2021

AMCA403S – RESOURCE MANAGEMENT TECHNIQUES

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. What are the components of linear programming problem?
2. What is a balanced transportation problem?
3. What is "no passing" rule in a sequencing algorithm?
4. Define Saddle point.
5. What is a critical path?

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Use Simplex method to solve the following LPP

   Maximize $z = 4x_1 + 10x_2$

   Subject to the constraints:

   $2x_1 + x_2 \le 50$

   $2x_1 + 5x_2 \le 100$

   $2x_1 + 3x_2 \le 90$

   $x_1 \ge 0 and x_2 \ge 0$

7. Solve the following assignment problem

	E	F	G	H
A	18	26	17	11
B	13	28	14	26
C	38	19	18	15

D	19	26	24	10

8. From the following information determine a sequence for the six jobs that will minimize the elapsed time.

Job	1	2	3	4	5	6
Machine A (in hours)	30	120	50	20	90	100
Machine B (in hours)	80	100	90	60	30	10

9. Solve the game whose payoff matrix is given by

PlayerB

$$\begin{array}{cc} & \begin{array}{ccc} B1 & B2 & B3 \end{array} \\ \begin{array}{c} \\ A1 \\ PlayerA \quad A2 \\ A3 \end{array} & \begin{pmatrix} -3 & -2 & 6 \\ 2 & 0 & 2 \\ 5 & -2 & -4 \end{pmatrix} \end{array}$$

10. Draw a network diagram for the following data

Activity	A	B	C	D	E	F	G	H	I
Preceding activities	-	-	-	A	A	B,D	C	B,G	F,G

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Use Big-M method to solve the following LPP

Minimize $z = 12x_1 + 20x_2$

Subject to the constraints:

$6x_1 + 8x_2 \geq 100$

$7x_1 + 12x_2 \geq 120$

$x_1 \geq 0$ *and* $x_2 \geq 0$

(or)

b) Use Graphical method to solve the following LPP

Maximize $z = 6x_1 + 20x_2$

Subject to the constraints: $2x_1 + 3x_2 \leq 30$

$3x_1 + 2x_2 \leq 24$

$x_1 + \leq 3$

$x_1 \geq 0$ and $x_2 \geq 0$

12. a) Obtain the optimal solution to the following transportation problem

	I	II	III	IV	V	Available
A	20	18	18	21	19	100
B	21	22	23	20	24	125
C	18	19	21	18	19	175
Demand	60	80	85	105	70	400

(or)

b) Solve the following assignment problem

	I	II	III
A	9	26	15
B	13	27	6
C	35	20	15
D	18	30	20

13. a) Use graphical method to minimize the time added to process the following jobs on the machines shown, i.e., for each machine find the job which should be done first. Also calculate the total time elapsed to complete both the jobs

Job 1	Sequence	A	B	C	D	E
	Time	3	4	2	6	2
Job2	Sequence	B	C	A	D	E
	Time	5	4	3	2	6

(or)

b) Use graphical method to minimize the time added to process the following jobs on the machines shown, i.e., for each machine find the job which should be done first. Also calculate the total time elapsed to complete both the jobs.

Job 1	Sequence	A	B	C	D	E
	Time	6	8	4	12	4
Job2	Sequence	B	C	A	D	E
	Time	10	8	6	4	12

14. a) Solve the following game graphically

$$\begin{array}{c} \\ PlayerA \end{array} \begin{array}{c} \begin{array}{cccc} B1 & B2 & B3 & B4 \end{array} \\ \begin{array}{c} A1 \\ A2 \end{array} \begin{pmatrix} 2 & 1 & 0 & -2 \\ 1 & 0 & 3 & 2 \end{pmatrix} \end{array}$$

(or)

b) Solve the following game

$$\begin{array}{c} \\ PlayerA \end{array} \begin{array}{c} \begin{array}{ccc} PlayerB & & \\ B1 & B2 & B3 \end{array} \\ \begin{array}{c} A1 \\ A2 \\ A3 \end{array} \begin{pmatrix} -3 & -2 & 6 \\ 2 & 0 & 2 \\ 5 & -2 & -4 \end{pmatrix} \end{array}$$

15. a) A project has the following time schedule

Activity	1-2	1-3	1-4	2-5	3-6	3-7	4-6	5-8	6-9	7-8	8-9
Time	2	2	1	4	8	5	3	1	5	4	3

Construct PERT network and compute total float for each activity and critical path and its duration.

(or)

b) The following information is given:

Activity		1-2	1-3	1-4	2-5	3-5	4-6	5-6
Estimated Duration	Optimistic	1	1	2	1	2	2	3
	Most likely	1	4	2	1	5	5	6
	Pessimistic	7	7	8	1	14	8	15

1) Draw the network
2) Find the expected duration and expected project length.
3) What is the probability that the project will be completed at least
4) Weeks earlier1han expected?

****